Our schema for Last Resort Hotels revolves around the room table. Each room has a number, and is located on a floor and in the wing of a specific building. Rooms also have various designations for smoking/non-smoking, whether it has a toilet and more. Key to our schema is the sleepingCapacity and seatingCapacity attributes in the room table: This implies whether the room can be used for sleeping, seating or both. Sleeping-only rooms have NULL for seatingCapacity; meeting-only rooms have NULL for sleepingCapacity; rooms used for both purposes have values for both attributes. Separate from sleeping versus meeting, each room has a type, such as suite, ballroom, pool room or other types. In the room_bed table, we assume that foldup and rollaway beds serve the same function (they are not permanent beds), and thus both are represented by a true value for isTemporary.

Sleeping and meeting creates different costs for the person paying for a room. sleepingRate and meetingRate are thus attributes in the room table, and can be NULL depending on the way(s) the room is used. meetingRate is a foreign key, though, because unlike sleepingRate — which the prompt states is fixed as a base rate) — meetingRate depends on the time of the day the room is reserved for (e.g. different rates for breakfast, lunch, dinner).

We assume that access cards can be lost, so both staff and guests can have multiple access cards throughout their time at the resort.

There are 3 kinds of charges that can be put in the records in the transaction_usage table. The rate for sleeping in a hotel room, the rate for using a meeting room which has different rates depending on hours of the day they are reserved for, and any other kind of service that can be charged for within the hotel complex, even if the customer is not a guest at the hotel. The transaction_usage table has one to many relationships. A one to many from room to transaction_usage, meeting_rate to transaction_usage, and service_rate to transaction_usage, referencing billing rates and bringing them into the transaction_usage table. Each charge is assigned to a billing party responsible for paying the charges. A singular charge can be allocated amongst more than one billing party. If an organization is paying for the use of a meeting room, the name of an authorized representative must be present in addition to the billing party to signify the involvement of an external organization.

A reservation is made by one billing party. The reservation table is the customer's preferences on the room, whereas the room_assignment entity is determined by the hotel so they can control for room-renovations and other various factors.

The hotel can host events. Events can take place in one or many rooms of the hotel. We **assume** that the host of an event is a single entity/person. As the project description, that entity/person can be a guest, billing party or both. Thus, the host table connects to both the guest and billing party tables, and the relationship depends on the identity of the host.